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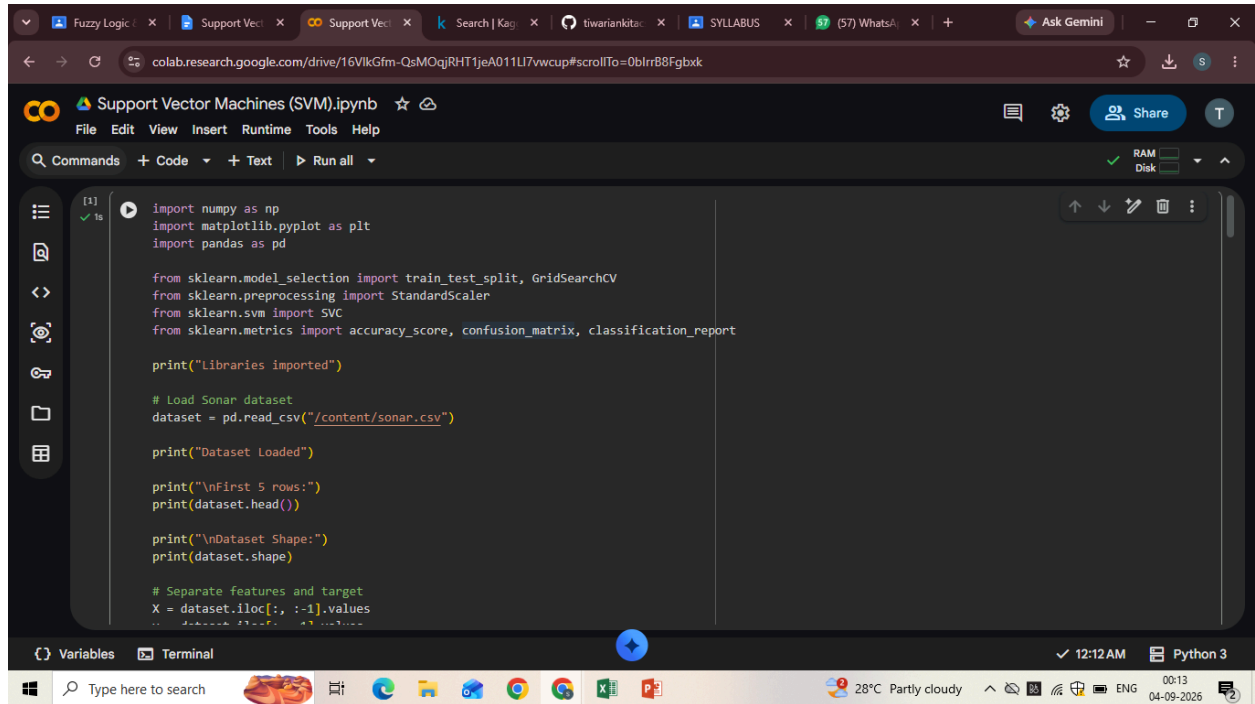
AIM:Support Vector Machines (SVM)

Implement the SVM algorithm for binary classification.

Train an SVM model using a given dataset and optimize its parameters.

Evaluate the performance of the SVM model on test data and analyze the Results.

INPUT:



```
[1] import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

print("Libraries imported")

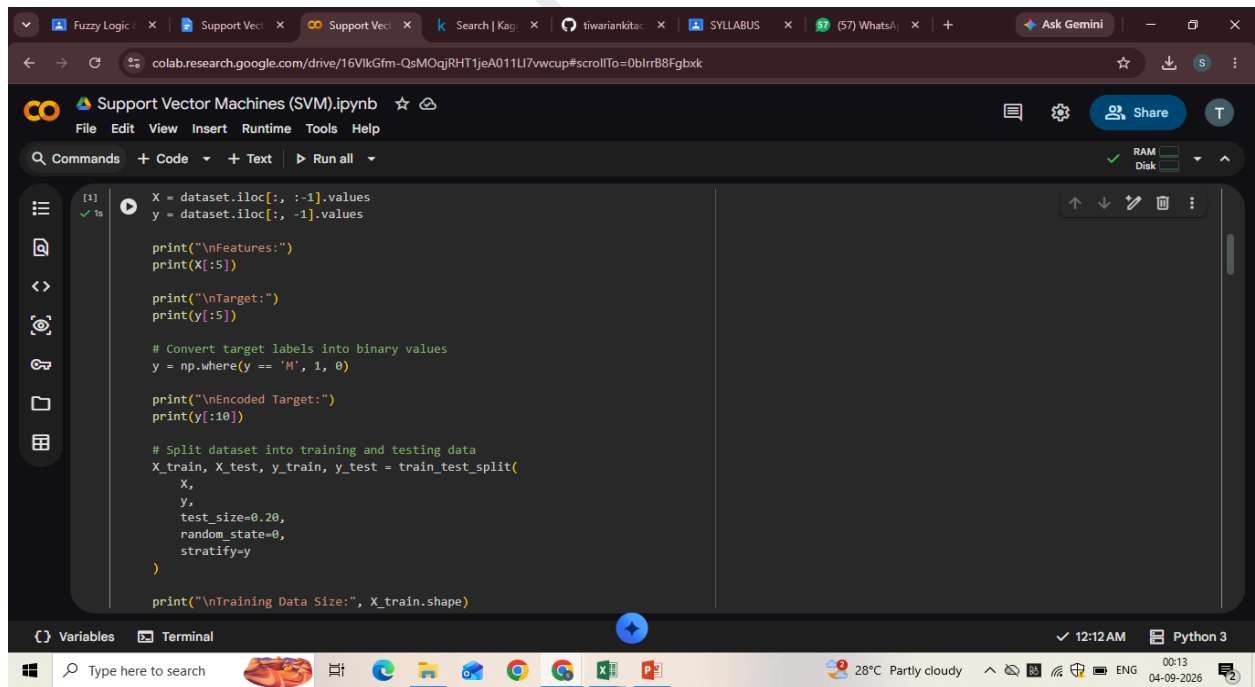
# Load Sonar dataset
dataset = pd.read_csv("/content/sonar.csv")

print("Dataset Loaded")

print("\nFirst 5 rows:")
print(dataset.head())

print("\nDataset Shape:")
print(dataset.shape)

# Separate features and target
X = dataset.iloc[:, :-1].values
```



```
X = dataset.iloc[:, :-1].values
y = dataset.iloc[:, -1].values

print("\nFeatures:")
print(X[:5])

print("\nTarget:")
print(y[:5])

# Convert target labels into binary values
y = np.where(y == 'M', 1, 0)

print("\nEncoded Target:")
print(y[:10])

# Split dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=0,
    stratify=y
)

print("\nTraining Data Size:", X_train.shape)
```

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```
[1] ✓ ts
print("Testing Data Size:", X_test.shape)

# Standardize the data
sc = StandardScaler()

X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)

print("\nData Standardized")

# Create SVM model
svm_model = SVC()

# Parameters for optimization
parameters = {
    'C': [0.1, 1, 10, 100],
    'kernel': ['linear', 'rbf', 'poly'],
    'gamma': ['scale', 'auto']
}

# Grid Search for best parameters
grid_search = GridSearchCV(
    estimator=svm_model,
    param_grid=parameters,
```

```
)

# Train and optimize SVM
grid_search.fit(X_train, y_train)

print("\nBest Parameters:")
print(grid_search.best_params_)

print("\nBest Cross Validation Accuracy:")
print(grid_search.best_score_)

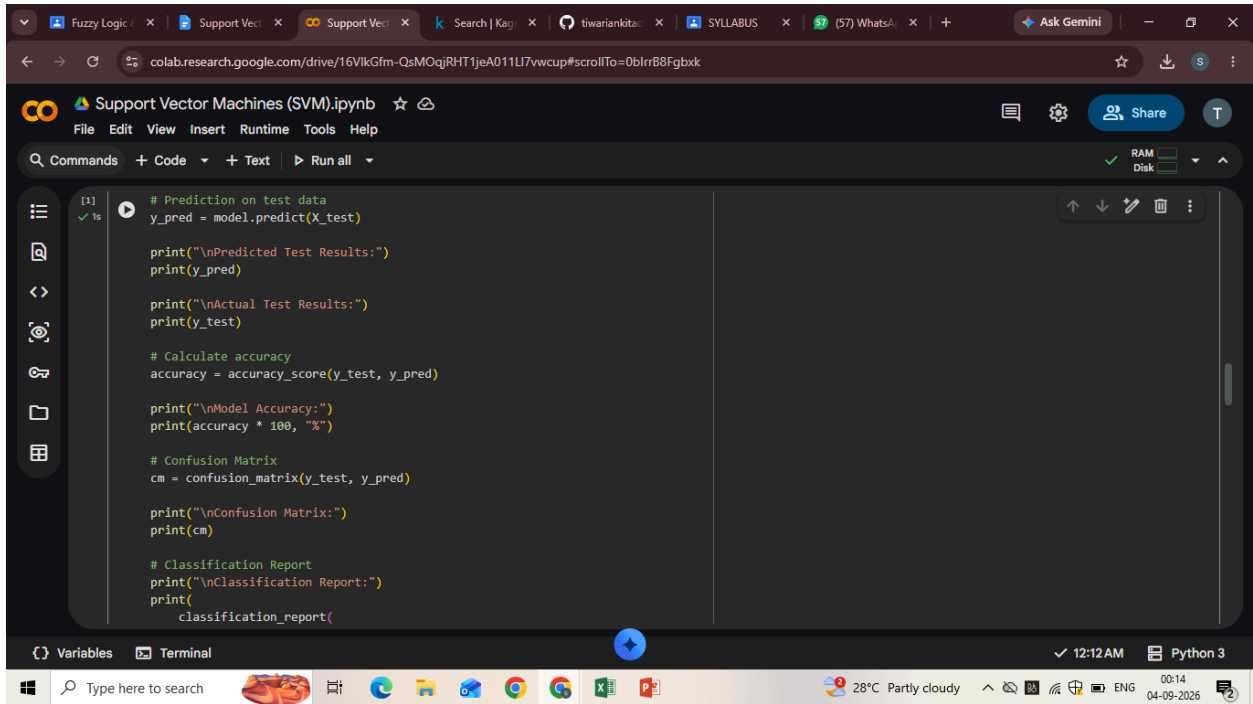
# Get best SVM model
model = grid_search.best_estimator_

print("\nBest SVM Model:")
print(model)

# Prediction on test data
y_pred = model.predict(X_test)
```

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```
[1] ✓ ts # Prediction on test data
y_pred = model.predict(X_test)

print("\nPredicted Test Results:")
print(y_pred)

print("\nActual Test Results:")
print(y_test)

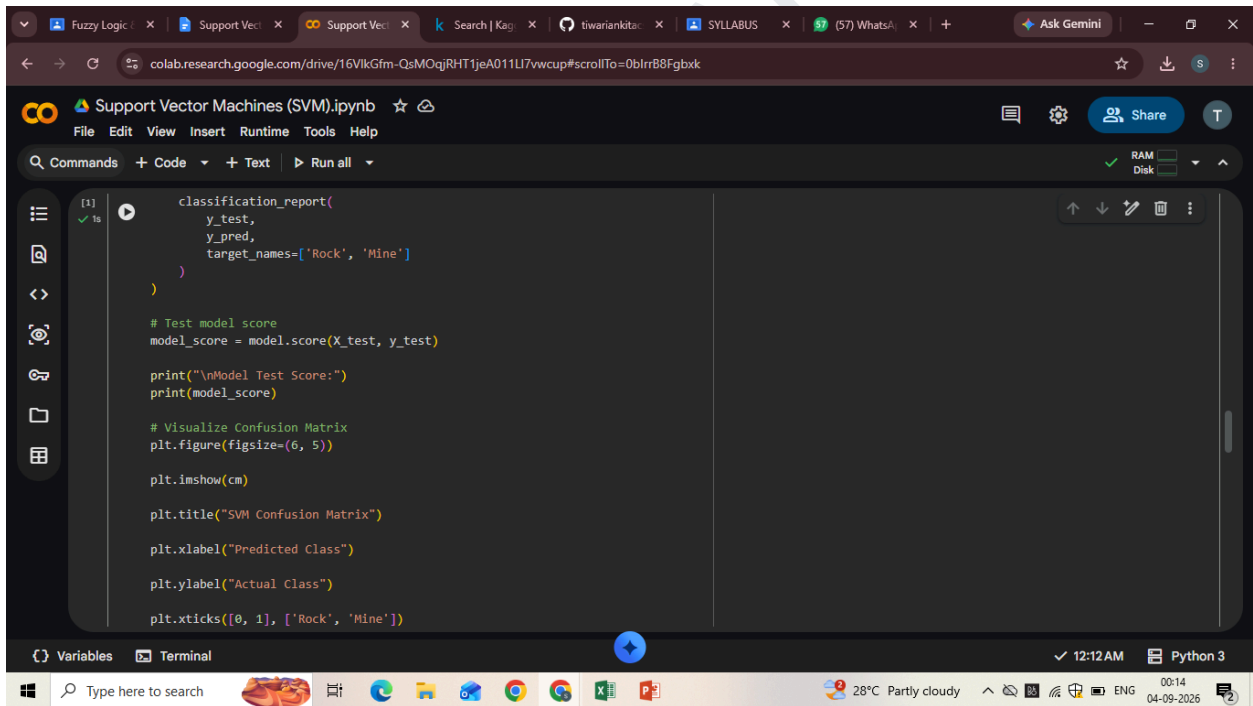
# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

print("\nModel Accuracy:")
print(accuracy * 100, "%")

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")
print(cm)

# Classification Report
print("\nClassification Report:")
print(classification_report(
```



```
)
    y_test,
    y_pred,
    target_names=['Rock', 'Mine']
)

# Test model score
model_score = model.score(X_test, y_test)

print("\nModel Test Score:")
print(model_score)

# Visualize Confusion Matrix
plt.figure(figsize=(6, 5))

plt.imshow(cm)

plt.title("SVM Confusion Matrix")

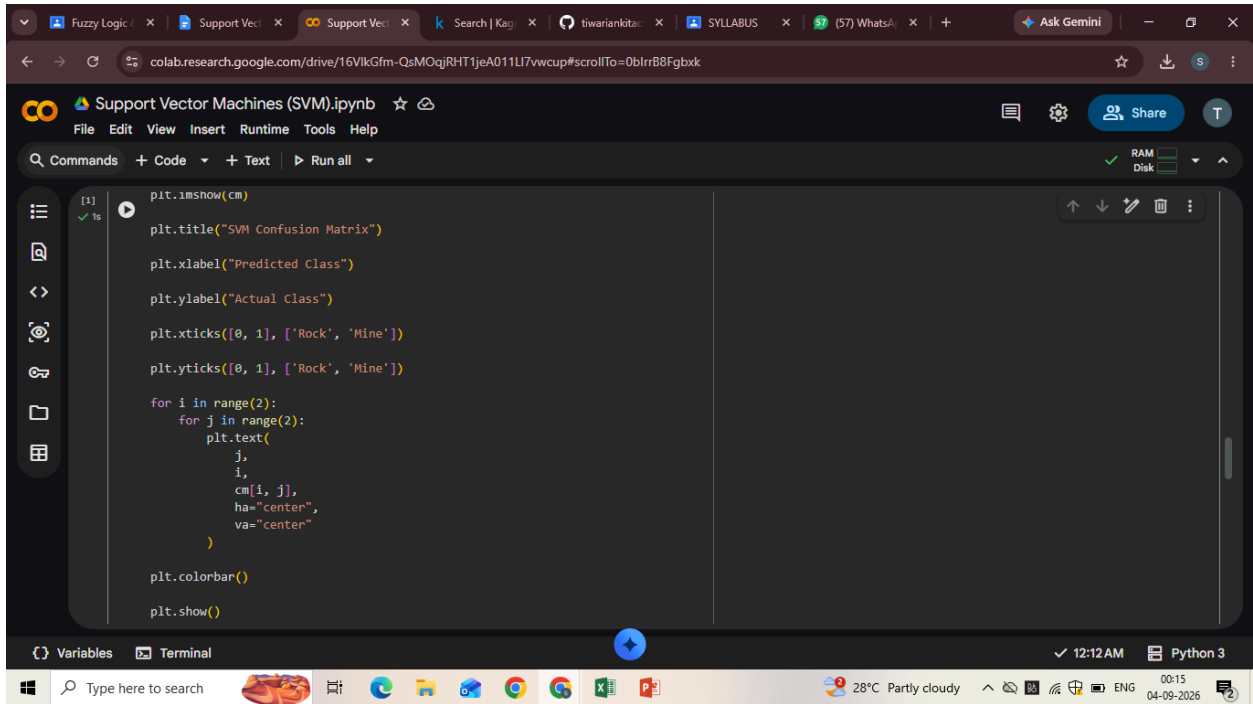
plt.xlabel("Predicted Class")

plt.ylabel("Actual Class")

plt.xticks([0, 1], ['Rock', 'Mine'])
```

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```
plt.imshow(cm)

plt.title("SVM Confusion Matrix")

plt.xlabel("Predicted Class")

plt.ylabel("Actual Class")

plt.xticks([0, 1], ['Rock', 'Mine'])

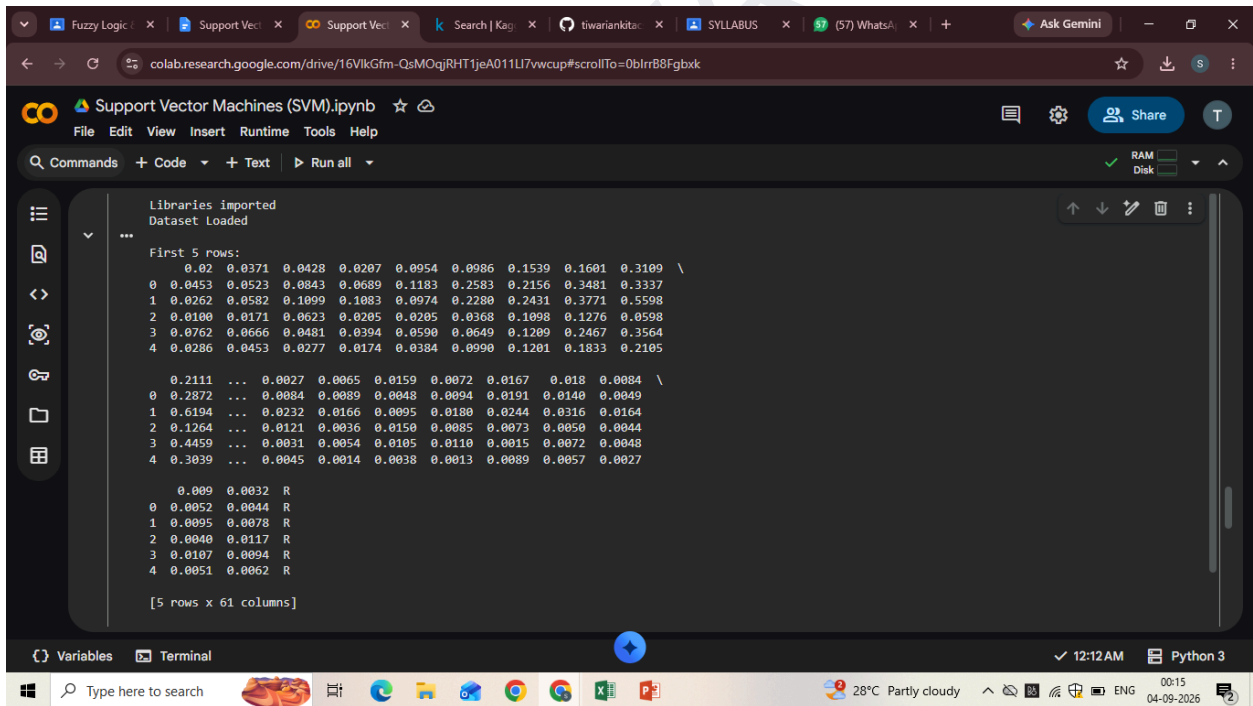
plt.yticks([0, 1], ['Rock', 'Mine'])

for i in range(2):
    for j in range(2):
        plt.text(
            j,
            i,
            cm[i, j],
            ha="center",
            va="center"
        )

plt.colorbar()

plt.show()
```

OUTPUT:



```
Libraries imported
Dataset Loaded

First 5 rows:
0.02 0.0371 0.0428 0.0207 0.0954 0.0986 0.1539 0.1601 0.3109 \
0.0453 0.0523 0.0043 0.0689 0.1183 0.2583 0.2156 0.3481 0.3337
1.0262 0.0582 0.1099 0.1083 0.0974 0.2280 0.2431 0.3771 0.5598
2.0100 0.0171 0.0623 0.0205 0.0205 0.0368 0.1098 0.1276 0.0598
3.0762 0.0666 0.0481 0.0394 0.0590 0.0649 0.1209 0.2467 0.3564
4.0286 0.0453 0.0277 0.0174 0.0384 0.0990 0.1201 0.1833 0.2105

0.2111 ... 0.0027 0.0065 0.0159 0.0072 0.0167 0.018 0.0084 \
0.2872 ... 0.0084 0.0089 0.0048 0.0094 0.0191 0.0140 0.0049
1.0194 ... 0.0232 0.0166 0.0095 0.0180 0.0244 0.0316 0.0164
2.0126 ... 0.0121 0.0036 0.0150 0.0085 0.0073 0.0050 0.0044
3.0459 ... 0.0031 0.0054 0.0105 0.0110 0.0015 0.0072 0.0048
4.0303 ... 0.0045 0.0014 0.0038 0.0013 0.0089 0.0057 0.0027

0.009 0.0032 R
0.0052 0.0044 R
1.0095 0.0078 R
2.0040 0.0117 R
3.0107 0.0094 R
4.0051 0.0062 R

[5 rows x 61 columns]
```

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The image displays two screenshots of a Google Colab notebook titled "Support Vector Machines (SVM).ipynb".

Top Screenshot: Shows the dataset shape and features. The dataset shape is (207, 61). The features are listed as a 20x60 matrix of numerical values.

```
[5 rows x 61 columns]
... Dataset Shape:
(207, 61)

Features:
[[0.0453 0.0523 0.0843 0.0689 0.1183 0.2583 0.2156 0.3481 0.3337 0.2872
  0.4918 0.6552 0.6919 0.7797 0.7464 0.9444 1. 0.8874 0.8024 0.7818
  0.5212 0.4052 0.3957 0.3914 0.325 0.32 0.3271 0.2767 0.4423 0.2028
  0.3788 0.2947 0.1984 0.2341 0.1306 0.4182 0.3835 0.1057 0.184 0.197
  0.1674 0.0583 0.1401 0.1628 0.0621 0.0203 0.053 0.0742 0.0409 0.0061
  0.0125 0.0084 0.0089 0.0048 0.0094 0.0191 0.014 0.0049 0.0052 0.0044]
[0.0262 0.0582 0.1099 0.1083 0.0974 0.228 0.2431 0.3771 0.5598 0.6194
  0.6333 0.706 0.5544 0.532 0.6479 0.6931 0.6759 0.7551 0.8929 0.8619
  0.7974 0.6737 0.4293 0.3648 0.5331 0.2413 0.507 0.8533 0.6036 0.8514
  0.8512 0.5045 0.1862 0.2709 0.4232 0.3043 0.6116 0.6756 0.5375 0.4719
  0.4647 0.2587 0.2129 0.2222 0.2111 0.0176 0.1348 0.0744 0.013 0.0106
  0.0033 0.0232 0.0166 0.0095 0.018 0.0244 0.0316 0.0164 0.0095 0.0078]
[0.01 0.0171 0.0623 0.0205 0.0205 0.0368 0.1098 0.1276 0.0598 0.1264
  0.0881 0.1992 0.0184 0.2261 0.1729 0.2131 0.0693 0.2281 0.406 0.3973
  0.2741 0.369 0.5556 0.4846 0.314 0.5334 0.5256 0.252 0.209 0.3559
  0.626 0.734 0.612 0.3497 0.3953 0.3012 0.5408 0.8814 0.9857 0.9167
  0.6121 0.5006 0.321 0.3202 0.4295 0.3654 0.2655 0.1576 0.0681 0.0294
  0.0241 0.0121 0.0036 0.015 0.0085 0.0073 0.005 0.0044 0.004 0.0117]
[0.0762 0.0666 0.0481 0.0394 0.059 0.0649 0.1209 0.2467 0.3564 0.4459
  0.4152 0.3952 0.4256 0.4135 0.4528 0.5326 0.7306 0.6193 0.2032 0.4636
  0.4148 0.4292 0.573 0.5399 0.3161 0.2285 0.6995 1. 0.7262 0.4724]
```

Bottom Screenshot: Shows the target variable, encoded target, training and testing data sizes, data standardization, best parameters, cross-validation accuracy, and predicted vs actual test results.

```
Target:
['R' 'R' 'R' 'R' 'R']

Encoded Target:
[0 0 0 0 0 0 0 0]

Training Data Size: (165, 60)
Testing Data Size: (42, 60)

Data Standardized

Best Parameters:
{'C': 10, 'gamma': 'scale', 'kernel': 'rbf'}

Best Cross Validation Accuracy:
0.8909090909090909

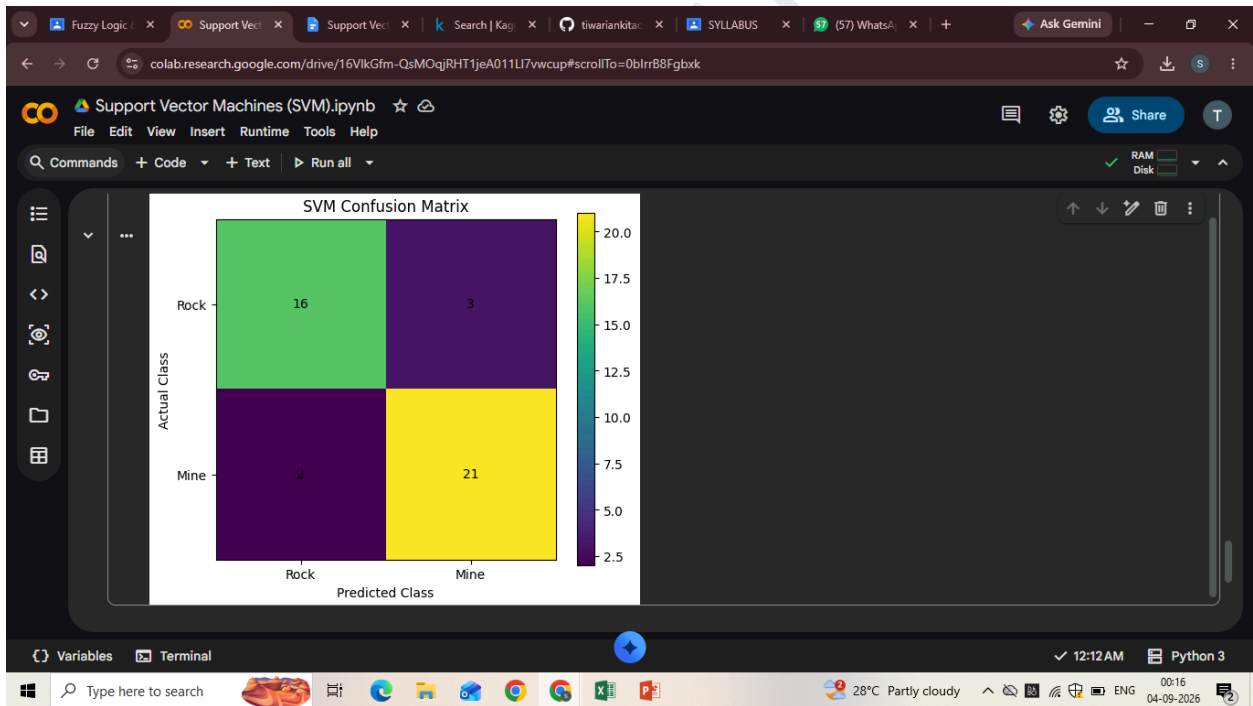
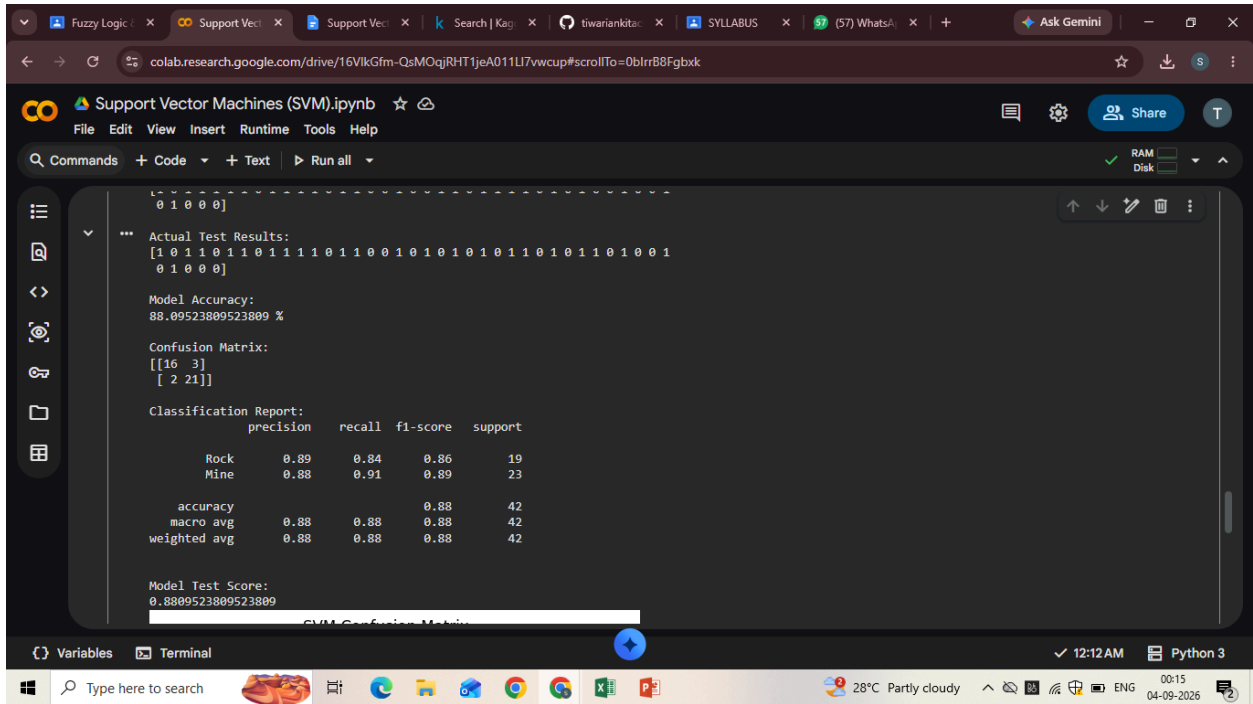
Best SVM Model:
SVC(C=10)

Predicted Test Results:
[1 0 1 1 1 1 1 0 1 1 1 1 0 1 1 0 1 1 1 1 1 0 1 0 1 0 0 1
 0 1 0 0 0]

Actual Test Results:
[1 0 1 1 0 1 1 0 1 1 1 1 0 1 1 0 1 0 1 0 1 1 1 0 1 0 1 0 0 1
 0 1 0 0 0]
```

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